ALGO TRADING

- 📈 **Leveraging Data & Machine Learning to Forecast Market Trends**
- 🔹 **What is Stock Price Prediction?**
- - Estimating future stock values using historical data, market indicators, and models.

- 🔹 **Why is it Important?**
- - Helps investors make informed decisions
- - Assists traders in identifying opportunities
- - Supports risk management strategies

- 🔹 **Approaches**
- - Statistical Models (ARIMA, Linear Regression)
- - Machine Learning (Random Forest, XGBoost)
- - Deep Learning (LSTM, Transformers)

- ✨ **Goal: Enhance decision-making by uncovering patterns hidden in financial data.**

SALIENT FEATURES

- 📊 **Trend Indicator (Buy / Hold / Sell)**
- 🔹 **Buy Signal**
- Price is in a **clear uptrend** (higher highs, higher lows).
- Moving averages (e.g., 50-day > 200-day).
- Technical indicators like **RSI < 30** (oversold) or **MACD crossover upward**.
- 🔹 **Hold Signal**
- Price is moving **sideways** (range-bound, no clear trend).
- Indicators show **neutral momentum** (RSI ~50, MACD flat).
- Investors prefer to wait for confirmation before action.
- 🔹 **Sell Signal**
- Price is in a **downtrend** (lower highs, lower lows).
- Moving averages (e.g., 50-day < 200-day).
- Technical indicators like **RSI > 70** (overbought) or **MACD crossover downward**.

INTERACTIVE DASHBOARD

Upload stock data (CSV or API)

- Visualize **price trends & moving averages**
- Auto-generate **trend signals (Buy / Hold / Sell)**
- Real-time charts with filters

RISK AND PROFIT CALCULATOR

Inputs:

- Entry Price, Stop Loss, Target Price
- Account Balance, Risk % per Trade, Leverage, Fees

- Outputs:
  - **Position Size** (shares/contracts)
  - **Risk Amount ($)**
  - **Profit/Loss scenarios**
  - **Risk-Reward Ratio (R:R)**
  - **Breakeven Price**

COMPARISON WITH CHOOSEN STOCK

- **Stock Selected: [Insert Stock Name / Symbol, e.g., AAPL]**
- **Actual vs Predicted**
- Actual prices taken from **historical data (last 6 months)**
- Predicted prices generated using **[your model: e.g., LSTM / ML + SMA]**
- Visual comparison through **line chart (Actual vs Predicted)**